

Family 1.0.tf: land and coast at ten kilometres and five days a dataset design note — the raw observations of the land surface, its coasts and its imagery

blauewelt/earth — design note

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Abstract

Family 1.0.tf (“tf” for *terrestrial, fine*) is the land-and-coast *family* — a numbered set of datasets built to one storage design and read by the programme’s forecasting model — of fine-granularity observation *stores* (a store is one dataset, published as a small set of arrays): measurements at **ten kilometres or finer and five days or finer** over the continents and a coastal band, on family 10’s (the programme’s granularity-aware dataset set) registry (the JSON file that lists every store), tiers (gridded fields G, point measurements P, image tiles T) and token (the unit the model reads: a value with its offset in space and time from the place being predicted and its footprint, the area and time span it averages over). It is designed for what a model of the land surface has to predict — where forest will be cleared, how much carbon a canopy holds and how fast that changes, where water will stand, what burns, what is planted, where cities grow — as well as for the forcing (the inputs that drive a system) the four ocean goals (El Niño, the ocean carbon sink, the Atlantic overturning circulation and sea-surface temperature) need from the land that borders the sea. The **land biosphere is the fifth read-out** beside those four, and it is scored on two held-out measurements: the carbon flux of flux towers (masts measuring the CO₂ a landscape takes up and gives back) the model was not trained on, and canopy height at laser footprints the model was not trained on. Three properties of the land record set the design. The fine record is mostly *points* (38,870 hourly and 132,437 daily weather stations, 2,932 radiosonde (weather-balloon) sites, 6,474 tide-gauge records, coastal buoys and radars, gliders (autonomous underwater profilers), flux towers (masts that measure the CO₂, water and heat exchanged between land and air), half a billion fire detections, and the LiDAR (laser ranging from orbit) footprints that measure vegetation in three dimensions), and they fit tier P (the point tier) with one widening of the time column. The daily *fields* at 0.05° are 20–30% valid and need the sharded, tiled, compressed layout family 1.gf (the global sibling at the same scale) introduces. And the *imagery* — optical at 10–30 m across forty years and four instruments on six satellites, radar at C and L band (wavelengths of about 5.5 and 24 cm) in dual and quad polarisation (two or four transmit–receive combinations), sub-metre orthophotos (map-corrected aerial photographs) where a country publishes them — is 0.01–3 petabytes a year per sensor and can never be pixels: it enters as a scene catalogue (a list of every image with its time, place and cloud cover, without the pixels) now and as cached local-codec tokens (compact vectors computed from the pixels by a small learned encoder) later, per the tier T (image tiles) of E-078, the programme’s multi-granularity design. Every candidate was checked against its producer on 2026-09-16; a licence rule sorts each store onto a public track (the public Hub repository — the project’s Hugging Face dataset store — readable by anyone) or a private track (built and held for this project’s own use, never republished), so a source’s terms decide where it lives and not whether it is used; and the note gives the stores — including the living surface: the lasers’ canopy structure, fluorescence and tower carbon as photosynthesis, leaf area and its calendar, biomass maps, and three billion species records — the arithmetic, a three-phase ladder (phase A ≈ 0.7 TB, and ≈ 1 TB[†] more for the biosphere wave), the falsifiers (the tests that can show a store is not worth keeping), and the decisions left to Chris.

1 Purpose — what the land family is for

The programme builds a forward model of the Earth’s surface state: encode what is observed at a place into a compact vector, predict it forward, read out anything (Chris, 2026-08-19). Families 7.2 (the global 0.25°, five-day input tensor) and 10 gave that model the ocean at 0.25° and the ocean’s point measurements; family 1.gf gives it the global fine scale. Nothing at ten kilometres or finer over land exists in any family yet, and the land is where the finest and most consequential predictions are asked for. This family exists for three uses, in this order.

Land change as a prediction target. Deforestation and forest degradation (where the next clearing happens, at 10–30 m, weeks ahead); the carbon a canopy holds and how it changes (biomass and canopy height, and the rate at which a forest can plausibly regrow after loss); surface water and floods; burned area; crop type and condition; urban growth. Each has an observed read-out today — alert systems at 10–30 m, annual loss maps back to 1990, LiDAR biomass, water-extent maps — and each is a fine-scale, land-only quantity that no ocean family can supply. Together they make the *land biosphere* the programme’s fifth read-out beside the four ocean goals, scored on two held-out measurements: the carbon flux of flux towers the model was not trained on, and canopy height at laser footprints the model was not trained on (§4.3).

The land as forcing for the ocean goals. The atmosphere over the continents that border the Atlantic and the Pacific was fitted by the reanalyses the model has seen so far; the stations, balloons, snow and land-temperature fields below are what those reanalyses were fitted to, and a model that has only seen the fit cannot learn where the fit fails (family 1.gf’s argument for ship reports, on land).

Coastal sea level and currents as read-outs of the AMOC and El Niño. Tide gauges since 1846 along the US east coast see the Florida Current; Pacific island gauges see Kelvin waves (equatorial waves carrying a sea-level bump eastward) weeks before Niño-3.4 (the standard El Niño index, the SST of a central Pacific box) moves; high-frequency radar measures the surface current at 6 km hourly where the boundary currents meet the shelf.

2 The rule, the two domains, and the two tracks — which sources qualify, land versus coast, and public versus private storage

The rule is family 1.gf’s: an observation with spatial support ≤ 10 km (a point counts) and temporal support ≤ 5 days; a derived product only where it carries raw observations otherwise inaccessible; coarser cadence only where critical, and named as an exception. Read-out *targets* are held to the spatial rule and not the temporal one — an annual loss map is what the model is asked to predict, not what it is shown. The biosphere wave (§4.3) names one exception beside E4–E6: **E7**, the eight-day leaf-area composite of MODIS and VIIRS (the imaging radiometers on NASA’s Terra and Aqua and on Suomi-NPP), stored as `lai500`. Eight days is three days coarser than the rule; it is admitted because leaf area and the fraction of sunlight a canopy absorbs have no finer global record, and the seasonal cycle they describe is what a forecast at weeks has to get right. Its 46 composites a year are carried as a `frame_table` (a list of each frame’s own dates), exactly as E4’s pentads are.

The two tracks. A source’s licence decides where its store lives, not whether it is built. A store whose terms allow redistribution (`licence.redistribution` of `yes` or `attribution`) goes on the *public track*: the public Hugging Face dataset repository, readable by anyone and by every agent that works on this project. A store whose terms forbid it (`no`) goes on the *private track*: the same nine arrays and the same registry entry, held in a private repository that only this project’s own jobs can read, used for training and evaluation here as a single licensee and never republished, forwarded or mirrored. The registry field `distribution` (`public` or `private`) records the track, and family 1.gf §4 gives the mechanism: where the private repository is, how a rented box reads it for one job, and what may be published from a model trained on it. Until 2026-09-16 the design had a licence *gate* that refused such sources outright, because every store was assumed to be republished; Chris removed the assumption the same day (“we can manage to build private datasets, and train on them without re-distributing”), and the five sources the gate had refused are re-sorted here. Two are set aside for reasons the rule itself gives: **HadISD** (the Met Office’s hourly station set; Non-Commercial Government Licence) is a quality-controlled derivative of the hourly reports `ghcnh` (this family’s hourly weather-station store) carries raw under CC0 (public domain), and **MSWEP** (a merged global precipitation product; CC BY-NC, non-commercial) is a merged product whose gauges are reachable through `ghcnd` (the daily weather-station store). Three become private-track stores: **ISMN** (the International Soil Moisture Network: 3,300 soil-moisture stations — “re-export or transfer of the original data to a third party is prohibited”) as `ismn`, the soil-moisture station store; **GRDC** (the Global Runoff Data Centre: 10,000 river gauges, no internet redistribution) as `river_grdc`, a private river-discharge store, beside the public `river`, whose CC BY 4.0 (free with attribution) Caravan subset — an open collection of catchment data — of 5,356 catchments stays public; and **ECA&D** (the European Climate Assessment & Dataset: 25,919 European stations, non-commercial, no redistribution)

as **ecad**, a private European station store, phase C and only for the series GHCN-Daily lacks. The same sorting applies inside a store: the research-only sources of **tide** (the tide-gauge store) within GESLA (the global tide-gauge compilation) — CMEMS, CV and UZ, three contributors’ codes — and the South African gauges are no longer dropped by a filter but written by the same build to a private sibling, **tide_private**; and the non-commercial tropical basemaps of Planet-NICFI (Planet’s monthly tropical mosaics, funded by Norway’s International Climate and Forests Initiative) join **cat_submetre** (the sub-metre orthophoto catalogue, §4.5) on the private track once their new-phase terms are read. One source stays out, on access rather than licence: **Dynamic World**, Google’s 10 m per-scene land-cover classification (CC BY 4.0), exists only inside Earth Engine, and an export at Sentinel-2’s scene volume would be a catalogue of a derived product.

Two domains. *Terrestrial*: the continents, inland water and ice sheets. *Coastal*: a band of ± 100 km around the GSHHG shoreline (the Global Self-consistent Hierarchical High-resolution Geography set, public domain), rasterised once at 0.05° into a static **dist_coast** (distance-to-coast) channel. For a gridded group “coastal” means the tiles whose minimum distance to coast is under 100 km; a point store holds every row it has and the consumer bounds by distance. The band is a build parameter recorded in the registry, not a property of the data.

What is held already and not repeated. Family 7.2’s land channels at 1° and its statics; family 10.1’s tropical moorings and along-track sea level; family 10.2’s daily 0.1° fishing effort (E-081, the experiment that built it), densest on the shelves.

3 Time axis, footprint and token — how every value is dated, sized and handed to the model

Time is family 10’s five-day bin (a *pentad*) from 1982-01-01, with the exact second in **time_s**; every row or pixel becomes the E-078 token — value, offset from the anchor in kilometres and days, and the two footprint fields $\log_2 f_p = \log_2(\text{support km}/27.83)$ and $\log_2 f_t = \log_2(\text{support days}/5)$. Family 1.gf’s conventions apply unchanged: -4 is the *label* for a point instrument with no areal support, a real support carries its true value even below -4 (a 30 m pixel is -9.9 , a 25 m LiDAR footprint -10.1 , an hour -6.9), and a coarse value is one token at its own centre, never copied into finer cells. Two land-specific points. A LiDAR footprint’s *vertical* axis is not a footprint field but a channel axis (§4.2). And an annual target map (a loss year, a biomass map) carries $\log_2 f_t = \log_2(365/5) = +6.2$ and is filed under the bin of its period’s end, so that a target is never read as an input from inside its own year.

4 The stores — every dataset, with its source, cadence, size and licence

Checked 2026-09-16 against producers’ pages, the NASA Common Metadata Repository, and live S3 (Amazon cloud storage) and directory listings; counts marked *measured* were summed from the producer’s inventory files; a dagger (†) marks an estimate. Bytes: a tier-P row costs $27 + 2C$ (schema 2) or $31 + 2C$ (schema 3, int64 **time_s**, for records starting before 1914); a sharded-grid pixel costs about $v \times 2C \times 1.2$ with v the valid fraction. In the *role* column, *forcing* is an input that drives the system and *read-out* a quantity the model is asked to predict; the *phase* is the build wave (A first).

4.1 Point networks, tier P — weather stations, radiosonde balloons, tide gauges, buoys, gliders, flux towers, fire detections, river gauges, soil-moisture probes, coastal radar and altimetry

store	source; channels (C)	points	cadence	record; rows	stored	licence	role; phase
ghcnd	<i>daily weather-station temperature, rain and snow.</i> GHCN-Daily (the Global Historical Climatology Network) v3.34 (NOAA NCEI, the US national environmental data centre): TMAX, TMIN, PRCP, SNOW, SNWD (maximum and minimum temperature, precipitation, snowfall, snow depth) ($C = 5$ channels, schema 3 — the time column widened to 64 bits for records before 1914)	132,437 stations, 42,984 active (measured)	daily	1763 $\rightarrow$; $\approx 1.0 \times 10^9$ station-days (3.2 B element rows)	≈ 40 GB	CC0	forcing; A
igra	<i>weather-balloon soundings of the air column.</i> IGRA (the Integrated Global Radiosonde Archive) v2.2 radiosondes (NCEI): T, dew-point depression, u , v (east and north wind), height at the 16 mandatory (standard pressure) levels ($C = 80$, schema 3)	2,932 stations, 807 active (measured)	00 / 12 UTC	1905 $\rightarrow$; 52.3 M soundings	10 GB	US government	forcing, El Niño; A
tide	<i>hourly coastal sea level from tide gauges.</i> GESLA-4 (4.1 download, July 2026) via the ERDDAP data server of UHSLC (the University of Hawaii Sea Level Center), 36 sources incl. UHSLC Fast Delivery and Research Quality, NOAA CO-OPS (NOAA's tide service): relative sea level ($C = 1$, schema 3)	6,474 records, 127,623 station-years	$\leq$ hourly	1800s $\rightarrow$ 2026; $1.2\text{--}1.5 \times 10^9$ on the public track	40–50 GB	mixed; the research-only sources go to tide_private	AMOC, El Niño; A
tide_private	<i>the tide gauges whose terms forbid republishing.</i> the GESLA-4 sources with research-only terms (CMEMS, CV, UZ) and, with the permission of SANHO (the South African Navy Hydrographic Office), the South African gauges; same schema as tide	the station-years the public sort removes [†]	$\leq$ hourly	as tide	10–20 GB [†]	research-only: private track	AMOC, El Niño; A (same build)
ndbc	<i>hourly weather and waves from US buoys and coastal stations.</i> NDBC (NOAA's National Data Buoy Center) buoys and C-MAN (coastal automated) stations (NOAA): wind u , v , gust, wave height, peak and mean period, direction, pressure, air T, SST ($C = 10$)	1,345 stations, 829 with 2025 data (measured)	hourly, 10-min recently	1970 $\rightarrow$ 2025; $3\text{--}5 \times 10^{8\dagger}$	15–25 GB	US government	SST, forcing; A (TAO sites — the tropical Pacific moorings — excluded: in gtmba , family 10.1's tropical-mooring store)
gliders	<i>profiles from autonomous underwater gliders.</i> IOOS Glider DAC (the US ocean observing system's glider data centre; 2,540 datasets) + OceanGliders GDAC (the international glider data centre; 1,120 deployments): T, S, O_2 , chlorophyll at the 16 Roemmich–Gilson pressures (the standard depths of the Argo climatology, 2.5 to 2,000 dbar) ($C = 64$)	$\approx 5 \times 10^6$ profiles [†]	profiles every ≈ 3 h	2003 $\rightarrow$	0.8 GB	open / CC BY 4.0	AMOC, SST, carbon; A
flux	<i>half-hourly land-air exchange of CO_2, water and heat, from towers.</i> FLUXNET Shuttle (the global flux-tower collection, 2026-04-30): eddy-covariance towers from AmeriFlux, ICOS, TERN, ChinaFLUX (the American, European, Australian and Chinese networks): NEE, GPP, Reco, LE, H, Rn, Ta, VPD, SWC, P (net ecosystem exchange, photosynthesis, respiration, latent and sensible heat, net radiation, air temperature, vapour-pressure deficit, soil water, precipitation) ($C = 10$)	700+ sites, $\approx 6,000$ site-years	half-hourly	1991 $\rightarrow$; $\approx 1.05 \times 10^8$	5 GB	CC BY 4.0 (six Tier-2 and Legacy sites excluded)	land carbon read-out; A

store	source; channels (C)	points	cadence	record; rows	stored	licence	role; phase
fire	<i>satellite detections of active fires</i> . FIRMS (NASA's Fire Information for Resource Management System) active fire: MODIS C6.1 1 km (2000-11 $\rightarrow$), VIIRS S-NPP / NOAA-20 / NOAA-21 375 m (2012 $\rightarrow$) (the imaging radiometers on NASA's and NOAA's satellites): brightness, FRP (fire radiative power), confidence, day/night ($C = 4$)	detections	per overpass	MODIS 1.1×10^8 , VIIRS $\approx 4 \times 10^8$ (2023 measured: 4.66 M / 21.75 M)	20 GB	NASA open [†]	read-out, carbon; A
river	<i>daily river discharge</i> . USGS NWIS (the US Geological Survey's water archive) daily discharge + GRDC-Caravan ($C = 1$, schema 3)	24,976 US sites, 8,691 active (measured); 5,356	daily	1800s $\rightarrow$; $\approx 4 \times 10^8$	14 GB	public domain / CC BY 4.0	read-out, freshwater; B
river_grdc	<i>daily river discharge worldwide (private)</i> . GRDC daily discharge, the full archive ($C = 1$, schema 3)	> 10,000 gauges, 160 countries [†]	daily	1800s $\rightarrow$; $\approx 1.5 \times 10^8$ [†]	$\approx 5 \text{ GB}^\dagger$	GRDC terms: private track	read-out, freshwater; B
ismn	<i>hourly in-ground soil moisture (private)</i> . ISMN (International Soil Moisture Network, TU Wien): soil moisture and soil temperature in up to five depth slots per station, the station's own depths kept in <code>platforms.json</code> , NaN where it has no sensor, with the network's quality flags ($C = 10$)	3,300 stations, 80+ networks	hourly	1952 $\rightarrow$; $\approx 3 \times 10^8$ [†]	$\approx 15 \text{ GB}^\dagger$	ISMN terms: private track	forcing; the in-situ truth for soil moisture; B
ecad	<i>daily European station weather (private)</i> . ECA&D (KNMI, the Dutch weather service) daily non-blended series — TX, TN, TG, RR, SS, SD, HU, PP (maximum, minimum and mean temperature, rain, sunshine, snow depth, humidity, pressure) ($C = 8$, schema 3) — for stations absent from GHCN-Daily only	subset of 25,919 series [†]	daily	1800s $\rightarrow$	$\leq 20 \text{ GB}^\dagger$	non-commercial, no redistribution: private track	forcing; C
ghcnh	<i>hourly weather-station reports</i> . GHCN-hourly v1 (NCEI), successor of the frozen ISD (Integrated Surface Database): T, dew point, pressure, u , v , precipitation, snow depth, cloud ($C = 8$, schema 3)	38,870 stations, 10,313 active (measured)	hourly / synoptic	1700s $\rightarrow$; 7.60×10^9 reports (measured)	360 GB	CC0	forcing; B
hfr	<i>hourly surface currents from shore-based radar</i> . HFRNet (the US high-frequency radar network) US (6 / 2 / 1 km) + European HFR Node (22 systems): u , v at the cell centre ($C = 2$)	$\approx 12.5 \text{ k}$ valid 6 km cells per hour, US	hourly	2012 $\rightarrow$ US, $\approx 2009 \rightarrow$ EU; $3.5\text{--}5 \times 10^9$ [†]	110–155 GB	free / CC BY 4.0	coastal currents, AMOC; B
alticap	<i>satellite sea level close to the coast</i> . ALTICAP coastal altimetry, Jason-3 (a radar altimeter satellite) 20 Hz, 0–500 km from coast (AVISO/ODATIS, French ocean-data services): anomaly, quality ($C = 2$)	along-track, $\approx 350 \text{ m}$	≈ 10 -day repeat	2016-02 $\rightarrow$ 2021-07 only	3 GB	CC BY 4.0	AMOC; B

Table 1: The point networks. Station stores ship a twelfth file, `platforms.json`, with each hash's source identifier, elevation and the source's station metadata (§6).

4.2 LiDAR footprints: the vertical structure of vegetation, tier P — laser ranging of forests and terrain from orbit

Optical and radar imagery see a canopy from above; only LiDAR measures it in three dimensions, and the two spaceborne instruments do so as sparse footprints along orbital tracks — exactly the case a point store was built for. The vertical axis goes into C the way pressure levels do for Argo and radiosondes: the footprint's own relative-height profile, not a single “height”.

store	source; channels (C)	footprint	record; rows	stored	licence	role; phase
gedi	<i>laser canopy height, cover and biomass from the Space Station.</i> GEDI (the Global Ecosystem Dynamics Investigation) on the ISS (International Space Station) (LP DAAC, NASA’s land data centre: L2A/L2B version 3; ORNL DAAC: L4A version 3, built from L2A version 3 and listed from 2019-04-04; v2.1 selectable): ground elevation, RH25, RH50, RH75, RH90, RH98 (the heights below which 25 to 98 % of the returned laser energy lies), canopy cover, plant-area index, footprint biomass and its error, and the shot’s sensitivity ($C = 11$; the quality and degrade flags go into qc)	25 m footprints every 60 m along 8 tracks, $\pm 51.6^\circ$; $\log_2 f_p = -10.1$	2019-04 $\rightarrow$ 2023-03 and 2024-04 $\rightarrow$ 2025-07 released (stored on the ISS between); “more than 10 billion high-quality observations”; L2A 106 TB native	$4\text{--}10 \times 10^9$ quality rows $\times$ 49 B $\approx$ 0.2–0.5 TB	EOSDIS open	canopy structure, biomass; A (was B) — one year, 2022, then the record on the probe’s numbers
icesat2	<i>laser terrain, canopy and sea-ice heights from a polar satellite.</i> ICESat-2 ATLAS (its photon-counting laser) (NSIDC, the US snow and ice data centre: ATL08, the land-and-vegetation product, release 7): terrain height, canopy height, canopy cover, photon counts per 100 m segment ($C = 6$); ATL07/ATL10 (its sea-ice products) sea-ice height and freeboard in the polar coastal band	≈ 11 m footprints at 0.7 m spacing, 6 beams, $\pm 88^\circ$; 100 m segments	2018-10 $\rightarrow$; ATL08 4.4–4.9 TB/yr native; segments $\approx 4 \times 10^9$ /yr over land [†]	$\approx 2.8 \times 10^{10} \times 39$ B ≈ 1.1 TB	NASA open	structure beyond 51.6° , ice; A (was C) — one year, 2022, then the record on the probe’s numbers

Table 2: The two orbital LiDARs. GEDI’s version 3 (2026-07-01) improved geolocation to about 10 m; the instrument’s status after July 2025 is not stated by the mission and is treated as ongoing. Phase A builds *one year* of each — 2022, a year both instruments flew in full — and the probe’s measured rows, bytes and fetch rate then decide how much of the rest is built and where; the sizes in the table are for the whole record.

4.3 The biosphere, tiers G, P and T — photosynthesis, leaf area and its calendar, canopy height and biomass maps, species occurrences, radar biomass

“Predict the biosphere” means, for this model, predicting what a named instrument would read at a place, a time and a footprint, and five things are asked of it. *Canopy structure* — how tall a forest is, how dense, how much carbon it holds — is measured directly only by the two orbital lasers of §4.2, and every canopy-height map below is a fit to them. *Photosynthesis*, the work the vegetation is doing while the satellite passes, is seen as solar-induced chlorophyll fluorescence (the faint red light a leaf re-emits while photosynthesising) from orbit and as carbon flux at the towers of **flux** (§4.1); the fluorescence is emitted by the process itself, and the towers are the long in-situ record, the biosphere’s equivalent of the moored overturning arrays. *Leaf state and its calendar* — how much leaf there is, and the dates of green-up and of senescence — are the seasonal cycle a forecast at weeks has to get right. *Disturbance* — where forest is cleared, what burns — is already carried by **lossyear**, **alerts**, **fire** and **burned500**. And *where life is* is the dated, georeferenced species records of GBIF (the Global Biodiversity Information Facility): the biosphere’s **icoads** (family 1.gf’s archive of ship weather reports, which carries the pre-satellite ocean) — sparse, biased towards birds and towards wealthy countries, and the only global record of species there is. None of it advects: leaf area, canopy height and species are where they are, so these channels’ drift is zero and their drivers are the shared atmospheric and soil channels, while their memory — weeks to months for leaf area, years to decades for canopy height — is longer than the model’s two-year cone. A slow channel is therefore read as a *state* and not as a history: last year’s map (canopy height, biomass, phenology, loss) enters as an input token at a lag of at least one year and the target is next year’s map at the same footprint, which is §3’s annual time footprint $\log_2 f_t = +6.2$ and not a new mechanism.

store	source; channels (C)	grid or footprint	record; rows or frames	stored	licence	role; phase; credential
canopy30	<i>the 2020 canopy-height map at 30m</i> . GLAD (the University of Maryland’s Global Land Analysis and Discovery laboratory) forest height 2020, from its Global Land Cover and Land Use Change 2000–2020 dataset (Potapov et al. 2022): Landsat metrics calibrated on GEDI’s relative heights; uint8 metres, and no water or no-data code: 0 is “no woody vegetation of 3 m or more”, so the map is dense ($C = 1$); annual sharded grid on the Hansen 10° tile grid cut into 2.5° groups exactly as lossyear	G · 30 m, annual (one bin)	2020; 261 tiles , 13–616 MB a tile (median 77), 35.95 GB in all, measured in the producer’s own directory on 2026-09-20; a strict subset of lossyear ’s 280	probe (tile 50N.000E, 2026-09-20): 281 MB stored for 304 MB fetched, valid fraction $1.0, 9.8\times$ over raw; ≈ 33 GB for the store, and 11–21 h for the 261 tiles in one bin, so a box	CC BY (the producer’s page names the Creative Commons Attribution licence and no version)	canopy read-out target; A ; no account — anonymous HTTP from GLAD’s transfer server
canopy_ref	<i>the finer canopy maps, catalogued; the pixels stay at the source</i> . ETH Zurich’s global canopy height 10 m 2020 (Lang et al. 2023: height and its standard deviation) and the Meta/WRI (Meta and the World Resources Institute) 1 m canopy-height maps (Tolan et al. 2024), cloud-optimised GeoTIFFs in the anonymous AWS bucket dataforgood-fb-data under forests/v1 and forests/v2 (listed 2026-09-20); one row per tile with its footprint, a sensor code and the asset URL, the tier-T catalogue form of §4.5	T · 10 m and 1 m	2020 (ETH; 3 TB native [†] , unverified — the host answered 429, “too many requests”, on 2026-09-20); 2018–2020 composite (Meta; tens of TB [†])	catalogue, < 1 GB	CC BY 4.0	reference for the canopy read-out; A for the catalogue, the pixels read by E-078c’s tile codec; no account
sif	<i>photosynthesis seen from orbit</i> . solar-induced chlorophyll fluorescence: TROPOMI (the trace-gas imager on Sentinel-5P) ungridded soundings at 740 nm (Caltech, Köhler et al. 2018) and ESA’s TROPISIF from the S5P-PAL service (the STAC collections L2_SIF_ and L2B_SIF_); fluorescence and its daily-corrected value, its $1\text{-}\sigma$ error, cloud fraction, solar and viewing zenith angle ($C = 6$); OCO-2 (v11.2r) and OCO-3 (v11r) SIF Lite from GES DISC (NASA’s atmosphere data centre) as second and third platforms with the same channels	P · 7×3.5 km (TROPOMI), 1.3×2.25 km (OCO); $\log_2 f_p = -2.49 / -4.02$ (geometric-mean support; the probe’s corner-derived median is -2.45); instantaneous	TROPOMI daily file 2018-05-01 → (2018-04-30 is the per-orbit product’s first day), OCO-2 2014-09 →, OCO-3 2019-08 →, all measured in the producers’ listings on 2026-09-20; Caltech’s 2018–2022 archive is 181.6 GB gzipped (measured); probe (2023-07, all 31 days): 101.7 M rows, 12.3 GB fetched in 577 s, no NaN, no out-of-bounds, i.e. 1.2×10^9 soundings a year	46.8 GB a year uncut (245 GB for 2018-05 to 2023-07, ≈ 390 GB to today); 63.9 % kept at cloud ≤ 0.2 ; the one daily file a day (0.3–0.4 GB) is the source, Caltech’s tarballs being 34 GB gzip streams with no member index ≈ 60 GB [†] per sensor-year	Caltech CC0; ESA Copernicus open; NASA open	photosynthesis read-out; A (TROPOMI now, OCO once GES DISC is approved); none for TROPOMI, Earthdata login plus GES DISC for OCO
lai500 (E7)	<i>leaf area and the light the canopy absorbs, every eight days (exception E7)</i> . MOD15A2H v061 (Terra), MYD15A2H v061 (Aqua) and VNP15A2H v002 (VIIRS on Suomi-NPP): leaf-area index, the fraction of photosynthetically active radiation the canopy absorbs, their standard deviations and quality ($C = 4$); the eight-day composite is the exception named in §2, carried as a frame.table exactly as chirps05 ’s pentads are	G · 500 m sinusoidal, sharded on the product’s own tile grid — the layout already holds a projected grid, family 1.gf’s polar-stereographic seaice.asi	Terra 2000-02-18 →, Aqua 2002-07-04 →, VIIRS 2012-01-01 → (measured, NASA’s Common Metadata Repository); 46 composites a year, 293 tiles a composite (measured: 13,466 Terra granules in 2020)	≈ 60 GB [†] per sensor-year	NASA open	leaf-state read-out, and forcing; A builds Terra 2020–2024 and measures, the record B; Earthdata login (in place)

store	source; channels (C)	grid or footprint	record; rows or frames	stored	licence	role; phase; credential
pheno500	<i>the leaf calendar, one map a year.</i> MCD12Q2 v061 land-surface phenology from Terra and Aqua together: the dates of green-up, mid-green-up, peak, senescence, mid-green-down and dormancy as day of year (int16), the minimum, amplitude and integral of the enhanced vegetation index, quality, and which growth cycle the row describes ($C = 10$)	G · 500 m sinusoidal, annual	2001 → 2025 (measured: 315 tiles for 2025, produced 2026-08)	$\approx 10 \text{ GB}^\dagger$ a year	NASA open	the calendar read-out; A ; Earthdata login
gbif	<i>three billion observations of organisms.</i> the GBIF occurrence snapshot, published monthly as Parquet (a columnar table format) in the anonymous AWS bucket <code>gbif-open-data-us-east-1</code> : every record with coordinates and a date to the day or the month; the time is the event date (noon where none is given, $\log_2 f_t = \log_2(1/5)$ for a day and $\log_2(30/5)$ for a month-precision record), the footprint the record's own coordinate uncertainty, platform the hashed taxon key, and the channels are kingdom code, class code, basis-of-record code and individual count ($C = 4$); <code>platforms.json</code> maps each hash to its taxon key and name; schema 3	P · point; every row carries its own footprint	1600s →; 3.67×10^9 records carry coordinates, no geospatial issue and a year, and the 2026-09-01 snapshot is 9,899 files and 285.3 GB — both measured on 2026-09-20	$\approx 90 \text{ GB}^\dagger$ on the public track	per record, measured: CC0 11.0 % and CC BY 72.2 % public; CC BY-NC 16.8 % private	where life is; A ; no account — anonymous S3
cat_ biomass_ esa	<i>ESA's P-band radar, catalogued.</i> the BIOMASS satellite (launched 2025-04-29), whose 70 cm wavelength penetrates a canopy to the trunks and is the one instrument that measures woody biomass directly from orbit: a catalogue of its released level-1 and level-2 products in the form of §4.5, if they are public	T	2025 →	catalogue	ESA terms, unverified : ESA's catalogue endpoints refused connections on 2026-09-20	reference for the biomass read-out; A if the products are public, else a phase-C row carrying the reason; to verify

Table 3: The biosphere stores, each with the credential its build needs. The canopy read-out's truth is the two orbital lasers of §4.2 and the photosynthesis read-out's is **flux**, the tower store of §4.1; disturbance is already carried by **lossyear**, **alerts**, **fire** and **burned500**. `canopy_ref` and `cat.biomass.esa` are catalogues in the sense of §4.5 — rows with asset URLs, no pixels copied.

4.4 Daily fields at 0.05° , tier G sharded — satellite land temperature, snow, surface colour, snow-and-ice charts and rainfall on a 5.6 km grid

store	source; channels (C)	grid	record; native	stored	licence	role; phase
1st05	<i>daily land-surface temperature, day and night.</i> MOD11C1 v061 (the MODIS product from NASA's Terra satellite) land-surface temperature day and night with QC (quality flags) ($C = 4$, $F = 5$ daily frames per bin); MYD11C1 (the same from Aqua) and VJ121C1 / VJ221C1 (from VIIRS on NOAA-20 2018 →, NOAA-21 2023 →) as continuation groups	0.05° , 7200×3600	Terra 2000-02-27 →; $\approx 45 \text{ MB/day}$, 9,591 granules	25.9 M px $\times$ 9,700 d, $v \approx 0.29 \times 0.5$: $\approx 350 \text{ GB}$ per satellite	NASA open	forcing, the land twin of SST; A (Terra), B (Aqua, VIIRS $\approx 350 + 160 \text{ GB}$)
snow05	<i>daily snow-cover fraction.</i> MOD10C1 v61 (MODIS snow cover; NSIDC): snow-cover fraction, cloud fraction as uint8 percent ($C = 2$); VNP10C1 / VJ110C1 continuation	0.05°	2000-02-24 →; 4–6 MB/day	$\approx 60 \text{ GB}$ (Terra; classes compress far below the float formula)	cite	forcing; A
ref105	<i>daily land-surface reflectance, the colour of the ground.</i> MOD09CMG v061 (MODIS reflectance on its 0.05° Climate Modeling Grid) daily surface reflectance bands 1–7 and state QA (cloud and quality flags) ($C = 8$); VJ109CMG / VJ209CMG continuation	0.05°	2000-02-24 →; $\approx 600 \text{ MB/day}$, 5.5 TB per satellite native	$v \approx 0.15$: $\approx 700 \text{ GB}$ (Terra), +330 GB VIIRS	NASA open	vegetation, snow, burn state; albedo; B

store	source; channels (C)	grid	record; native	stored	licence	role; phase
ims1k	<i>daily analyst-drawn snow and ice chart</i> . NOAA IMS (the Interactive Multisensor Snow and Ice Mapping System) daily Northern-Hemisphere snow and ice chart (NSIDC G02156), 1 km class (uint8, $C = 1$); 4 km from 2004, 24 km from 1997	24024 ² polar stereographic	1 km 2014-12-02 →; 2.8 MB/day gz	classes compress $\approx 50\times: \approx 50$ GB	cite	forcing; B
chirps05 (E4)	<i>five-day rainfall from satellite infrared and rain gauges (exception E4)</i> . CHIRPS (Climate Hazards Center InfraRed Precipitation with Station data) v3.0 pentad precipitation (UCSB CHC, the Climate Hazards Center at UC Santa Barbara): infrared + climatology + gauges, land 60°S–60°N ($C = 1$); its six-per-month pentads carried as a frame_table (a list of each frame's dates), each frame filed under the family bin holding its midpoint	0.05°, 17.3 M px	1981 →; $\approx 3,300$ pentads	$v \approx 0.3: \approx 40$ GB	CC BY 4.0	forcing, teleconnection read-out; B

Table 4: The daily fields. Terra and Aqua are still producing (end late 2026 / 2027, date not fixed); Suomi-NPP data end 2026-11-01, so the VIIRS continuation is NOAA-20 and NOAA-21; MODIS Collection 7 replaces v061 when published.

4.5 The image catalogue, tier T — every satellite and aerial image listed by time, place and cloud cover; the pixels stay at the source

Every source below is 0.01–3 petabytes a year; none is stored as pixels. Each is a tier-P *catalogue* store: one row per scene-date with **bin**, **time_s** (acquisition), **lat**, **lon** (footprint centre), **platform** (the hashed scene identifier), **qc** (the producer's processing baseline), **fp** (the tile's footprint; a 110 km Sentinel-2 tile is +2.0), and $C = 5$: cloud fraction, valid fraction, sun zenith or radar incidence, area, and a **sensor code** — which instrument on which platform, from a code table in **store.json**. A sidecar **assets.parquet** (a table file) maps the hash to the STAC identifier (the scene's name in the producer's SpatioTemporal Asset Catalog) and asset URLs, so the token builder of E-078c (the first experiment to train the local image codec, on about 300 coastal cells) can fetch exactly the scenes a cell needs. The token cache itself (E-078 §4: k tokens of 64 dimensions per tile-date from a local codec conditioned on the sensor code and its geometry) is a later tier-P store whose size E-078c measures. Two more catalogues are built in the same form and listed with the biosphere stores instead of here, because what they catalogue is canopy and biomass rather than raw imagery: **canopy_ref**, the 10 m and 1 m canopy-height maps, and **cat_biomass_esa**, the products of ESA's P-band BIOMASS satellite (§4.3).

store	source	scenes	record; native volume	licence	note
cat_landsat	<i>every Landsat scene since 1982: 30 m optical and thermal images</i> . Landsat Collection 2 Level 2, all sensors : Landsat 4 and 5 TM (1982 / 1984 → 2012), 7 ETM+ (1999 → 2024), 8 and 9 OLI/TIRS (2013 / 2021 →) (TM, ETM+, OLI and TIRS are the successive Landsat cameras); 30 m reflectance, 60–120 m thermal (USGS)	$\approx 1,400$ – $1,500$ per day today [†] ; $\approx 10^7$ in the archive	1982 →; 0.3–0.5 PB/yr today [†]	public domain	the forty-year record, one sensor code per row
cat_hls	<i>NASA's harmonised Landsat and Sentinel-2 scenes</i> . HLS (Harmonized Landsat Sentinel-2) v2.0 S30 + L30 (from Sentinel-2 and from Landsat), 30 m harmonised Sentinel-2 and Landsat 8/9 (LP DAAC)	2.71 M + 1.75 M granules in 2025 (measured)	2013-04 / 2015-11 →; ≈ 0.7 PB/yr	NASA open	the harmonised sample of the same record
cat_s2	<i>every Sentinel-2 scene: 10 m optical images from the EU</i> . Sentinel-2 L2A (surface reflectance), 10 / 20 / 60 m; S2B + S2C (S2A retired 2026-03) (ESA / CDSE, the Copernicus Data Space Ecosystem)	12.7–15.5 k per day (measured), ≈ 600 MB each	2015-07 →; ≈ 3 PB/yr	Copernicus open	CDSE account; 12 TB per 30 days
cat_s1	<i>every Sentinel-1 scene: 10 m C-band radar images from the EU</i> . Sentinel-1 IW GRDH (the wide-swath mode, ground-range detected, high resolution) dual-pol VV/VH (HH/HV at high latitude) (vertical or horizontal transmit and receive), 10 m pixels; S1C + S1D (S1A retired 2026-06-30); the SLC (single-look complex, the form that keeps the radar phase) is catalogued beside it for the dual-pol covariance	982 per day (measured), 1.72 GB	2014-10 →; GRD ≈ 620 TB/yr, SLC ≈ 1.6 PB/yr	Copernicus open	C-band polarimetry: dual, never quad

store	source	scenes	record; native volume	licence	note
cat_nisar	<i>every NISAR scene: L-band radar images, 2026 onward.</i> NISAR (the NASA-ISRO radar satellite) L2 GCOV provisional (2026-07-20 →): L-band geocoded covariance, quad-, dual- and compact-pol modes, 10 / 20 / 80 m posting (ASF, the Alaska Satellite Facility)	28,456 granules in August 2026	2026-06-17 →; 0.7–1 PB/yr	NASA open [†]	the only free quad-pol L-band
cat_palsar (E5)	<i>Japanese L-band radar mosaics, yearly (exception E5).</i> JAXA (Japan's space agency) PALSAR / PALSAR-2 global 25 m mosaics v2.6 (HH, HV; 1° tiles) and the ScanSAR (wide-swath mode) L2.2 scenes	≈ 15 k tiles per year	mosaics 2007–2010, 2015–2025; ScanSAR 2014-08 →	JAXA research terms, ©JAXA, registration	annual: an exception, the L-band record before NISAR, and the radar half of the biomass read-out (§4.3). Phase A (was B), conditional on a free JAXA G-Portal account (the registration its terms require), which is Chris's to open; without it the store waits coastal tokens
cat_olci	<i>300 m coastal ocean-colour scenes.</i> Sentinel-3 OLCI (Ocean and Land Colour Instrument) L2 WFR (water, full resolution) 300 m, coastal band; GlobColour (Copernicus' merged ocean-colour product) OLCI 300 m L3 (Copernicus 009_103) as the gridded form, which phase C may also store as a sharded tier-G group over the coastal band (oc300.coast, chlorophyll and flags, ≈ 1.6 TB)	457 per day, 150 GB/day (measured)	2016-04 →; ≈ 55 TB/yr	Copernicus open	
cat_viirs	<i>375 m VIIRS imager scenes.</i> VIIRS I-band L1B (the imaging bands of the Visible Infrared Imaging Radiometer Suite, as calibrated radiances) 375 m (VNP02IMG / VJ102IMG, LAADS, NASA's atmosphere archive)	225–242 granules/day	2012 / 2018 →; ≈ 15 TB/yr per satellite	NASA open	
cat_ecostress	<i>70 m thermal images from the Space Station.</i> ECOSTRESS (the ECOSystem Spaceborne Thermal Radiometer) L2T LSTE (tiled land-surface temperature and emissivity) 70 m thermal from the ISS, irregular	1.37 M granules in 2024	2018-07 →; ≈ 10 TB/yr	NASA open	no cadence; catalogued only
cat_submetre (E6)	<i>sub-metre aerial photographs where a country publishes them (exception E6).</i> sub-metre and metre-scale orthophotos where a country publishes them openly: USDA NAIP (the US Department of Agriculture's aerial imagery programme) 0.6–1 m (US, 2–3-yearly, public domain), swisstopo (Switzerland's federal mapping office) SWISSIMAGE 10 cm (Switzerland, 3-yearly, open government data); Planet-NICFI 4.77 m monthly tropical basemaps (non-commercial; programme contract ended 2025-01, basemaps in Earth Engine to 2026-08, new-phase terms unverified; private track)	regional	2009 / 2017 / 2015 →	public domain / OGD / non-commercial (NICFI: private track)	regional, multi-year: an exception; no free sub-metre source is global or daily

Table 5: The catalogued imagery. The ten catalogues together are $\approx 1.6 \times 10^8$ rows and ≈ 7 GB; pixels are never copied.

4.6 Read-out targets — what the model is asked to predict: forest loss, burned area, floods, biomass, land cover

Targets are what the footprint-queried decoder (E-078 §5: the part of the model that answers a question about a given place and area) is asked to predict at the target's own footprint. They are held to the spatial rule only. One further target is not repeated here: `canopy30`, the 2020 canopy-height map at 30 m, is a read-out target in exactly this sense and has its row in the biosphere table (§4.3), beside the lasers it was fitted to.

store	source; channels	resolution	record	stored	licence	phase
alerts	<i>dated forest-clearing alerts.</i> forest-disturbance alerts as point rows (date, position, confidence, system): GLAD-L (the University of Maryland's Landsat alerts; 30 m, tropics, 2015 →), GLAD-S2 (its Sentinel-2 alerts; 10 m, 2019 →), RADD (Radar for Detecting Deforestation; Sentinel-1 10 m, 6–12 day, humid tropics, WUR — Wageningen University —, CC BY 4.0), OPERA DIST-ALERT (NASA OPERA's disturbance alerts; 30 m global, 2023 →)	10–30 m	2015 →; rows ≈ 10 ^{9†}	≈ 30 GB	CC BY 4.0 (GLAD terms unverified)	A
lossyear	<i>the year each 30 m pixel lost its forest.</i> Hansen Global Forest Change (the University of Maryland's annual forest-loss map) v1.13 tree cover 2000 and loss year 2001–2025 (30 m, CC BY 4.0); JRC Tropical Moist Forest (the EU Joint Research Centre's tropical forest-change map) v2025 annual change 1990–2025 (30 m, free without restriction); annual uint8 sharded grids	30 m	1990 / 2001 → 2025	loss is sparse: ≈ 30 GB	CC BY 4.0 / free	A
burned500	<i>the date each 500 m pixel burned.</i> MCD64A1 (the MODIS burned-area product) v6.1 burned-area date (500 m, monthly, LP DAAC)	500 m	2000–11 →	≈ 10 GB	unrestricted	A
cat_gfm	<i>catalogue of flood and surface-water maps.</i> Copernicus Global Flood Monitoring; Sentinel-1 flood extent per pass at 20 m, archive to 2015, ≈ 2 M scenes; OPERA DSWx-HLS / DSWx-S1 (NASA OPERA's dynamic surface-water extent from HLS and from Sentinel-1) 30 m water extent	20–30 m	2015 / 2016 →	catalogue, < 1 GB	free (GFM terms unverified) / NASA open	A
biomass100	<i>yearly above-ground biomass maps, 100 m.</i> ESA CCI (ESA Climate Change Initiative) Biomass v7.0 (2026-05-21): above-ground biomass and its standard deviation, 100 m, years 2005–2012 and 2015–2024 (measured in the producer's directory on 2026-09-20); GEDI L4B (GEDI's gridded biomass) 1 km biomass 2019-04 → 2023-03 (measured)	100 m	annual	760 GB native; ≈ 0.4 TB	ESA CCI terms, read 2026-09-20: any user, any purpose, with acknowledge- ment; redistribution unaddressed (decision 12)	A for 2020 and for the GED I L4B group (was C); the other years B
landcover10	<i>yearly land-cover and crop-type maps.</i> annual 10 m land cover as an annual target: Esri / Impact Observatory 2017–2025 (CC BY 4.0), ESA WorldCover 2020 and 2021 (CC BY 4.0), MapBiomas (Brazil's annual land-cover project) Brazil 1985–2024 (30 m, CC BY 4.0); crops: USDA CDL (Cropland Data Layer) 30 m annual (public domain), EUCROPMAP (the JRC's EU crop map) 2018 / 2022, WorldCereal (ESA's global crop map) 2021	10–30 m	annual	≈ 0.2 TB [†]	CC BY 4.0	C

Table 6: The read-out targets. GHSL (the EU's Global Human Settlement Layer) built-up (100 m, five-yearly, 10 m for 2018) and WSF Evolution (the World Settlement Footprint time series; 30 m annual 1985–2015, CC BY 4.0) are the urban targets and join **landcover10** in phase C. The three annual targets of the living surface are **canopy30** (how tall the canopy is), **pheno500** (when it leafs out and when it dies back) and **lossyear** (where it was cleared) — structure, calendar and clearing, one map a year each.

4.7 Statics — fixed maps: elevation and sea-floor depth, land-cover fractions, distance to coast

static_fine (the store of fixed fields): GEBCO.2026 (the General Bathymetric Chart of the Oceans) elevation and bathymetry at 15'' (7.5 GB as float16, public domain); WorldCover 2021 (ESA's 10 m land-cover map) class fractions at 0.05° (11 channels, 0.6 GB, CC BY 4.0); **dist_coast** at 0.05° from GSHHG; and, for the 10 m tile tier, Copernicus DEM GLO-30 (the 30 m Copernicus elevation model) referenced on AWS rather than copied. About 8 GB, phase A.

5 Missing values and quality — how gaps and bad values are recorded

A channel not measured is NaN (not-a-number); a value outside its physical bound is NaN and counted, never clipped; the source’s own quality flag is kept in `qc` and nothing is homogenised. Fields carry their quality layers as channels (`1st05`’s QC day and night; `ref105`’s state QA) rather than as a mask, so the consumer decides the threshold. Absence is information: a station that stopped, a LiDAR track that was not flown, a cloudy day — all read as $n_R = 0$ (no readings in the token) or a NaN pixel, and the model is told so.

6 Design decisions — the choices made, and why

Licence decides where a store lives, not whether it exists. The first draft of this note refused four of the best land networks because every store was to be republished, and accepted having no soil-moisture store as the price. The two-track rule of §2 removes the price: a source that may not be redistributed is built once, held privately, used here as a single licensee, and never leaves. The cost that remains is named in family 1.gf §4 — a private store is invisible to the other agents and to anyone reproducing the work — so a result that depends on one says so, and the public track stays the default wherever a licence allows it.

Vertical structure is stored natively, as the LiDAR’s own profile. A canopy is a three-dimensional object and imagery cannot see into it. `gedi` therefore keeps the footprint’s relative-height profile (RH25 to RH98), cover and plant-area index as channels, and `icesat2` its terrain and canopy heights per segment — the vertical axis in C , as depth is for Argo. A single “canopy height” derived product would discard exactly what the instrument was flown for.

State and event are both stored, so that a rate is measurable. A forest can be cleared in a day and takes decades to regrow; biomass and height change slowly and within physical limits. The family keeps the *state* (height and biomass from LiDAR, annual biomass and land-cover maps) and the *event* (dated alerts, loss years, burned-area dates) as separate stores with their own timestamps, so that a model’s predicted change can be checked against what growth and loss can physically do in the interval. Whether such bounds are enforced in the model or only measured is the model’s decision; the data design makes the measurement possible.

One catalogue across forty years of sensors, never blended. `cat_landsat` carries Landsat 4, 5, 7, 8 and 9 from 1982 with a sensor code per row, and `cat_hls` carries NASA’s harmonised sample of the recent part; the two coexist. The catalogue does not blend or calibrate across sensors — that is the local codec’s job, conditioned on the sensor code and geometry (E-078 §4), and the falsifier in §9 asks whether it learned the shift. Radar is stored the same way: C-band dual-pol from Sentinel-1 (there is no quad-pol mode), L-band dual-pol from PALSAR and dual- and quad-pol from NISAR, each under its own code.

The catalogue before the codec. Ten catalogues at 7 GB turn the tile tier’s scene counts into measured numbers before a single token is computed, and are the only way to size E-078c against reality; they are phase A even though the tokens wait.

The coastal band is a build parameter. ± 100 km covers the shelf break almost everywhere and the radars’ 200 km reach; it is recorded in the registry so a rebuild at another width is a rebuild, not a redesign.

Station elevation lives in the platform table. Hashing a station identifier would lose the one number that makes a 2 m temperature comparable with a 0.05° skin temperature; `platforms.json` (the per-station metadata file) keeps it, with the source’s own metadata, beside the nine arrays.

Annual products are exceptions, and named. E4 CHIRPS (pentad, derived, gauge-bearing, CC BY 4.0) is the one derived field admitted as an input; E5 PALSAR mosaics (annual, the L-band record before NISAR) and E6 regional sub-metre orthophotos (multi-year cycles) are admitted to the catalogue only; E7 the eight-day MODIS and VIIRS leaf-area composite (`lai500`, three days coarser than the rule,

the only global record of how much leaf there is) is admitted as an input and carries its composites' own dates in a `frame_table`, as E4 does. Targets are outside the temporal rule by construction.

What the imagery is for. The catalogue lists optical at 10–30m across six sensors, radar in two polarisations at two bands, LiDAR footprints and sub-metre orthophotos where they exist — the raw inputs a land representation must be learned from. Family 0.9.tf (this family with its imagery replaced by Google's AlphaEarth embedding) is the alternative that takes a published embedding — computed from the same optical and C-band imagery and trained against the same LiDAR, L-band and climate sources — in place of the pixels, and inherits every other store of this family by reference; the two families are meant to be compared, not merged.

7 Validation — the checks every store must pass

Every array's sha256 (file checksum) against `store.json`; N from the file sizes; `bin` monotone and equal to `floor_divide(time_s, 432000)`; the CSR (compressed-sparse-row bin) index reproduced; for sharded grids, every index entry decompressing to a tile of the declared shape and NaN fraction, and per-year frame counts equal to the producer's listing; for catalogues, scene counts per day against the producer's own catalogue; for `tide`, the station-years sorted to `tide_private` reported. A day absent upstream is a zero-length frame with a counter, never a silent gap; a fetch lane refuses an empty listing or a short download and records any relaxation in `degraded` (the build's list of relaxed checks).

8 Limitations — what is unverified, or cannot be done

`hfr`'s archive size is unverified because the UCSD (University of California San Diego) server refused connections on the day of checking. GEDI's operating status after July 2025 is not stated by the mission. `icesat2` at native segments is a terabyte, so phase A builds one year of it and the probe's numbers decide the rest; a thinned version would be derived. ETH's 10m canopy-height volume is unverified because that host answered “too many requests” on the day of checking, and ESA's BIOMASS access terms are unverified because its catalogue endpoints refused connections the same day. The sub-metre tier is regional and multi-year by the nature of what is published openly. The tile tokens depend on a codec that does not yet exist (E-078c); until it does, the tile tier is a catalogue. Dynamic World cannot be built for lack of a bulk copy. A private-track store can be checked by nobody outside the project, so §7's validation is the only certificate it has, and a published result that rests on one names it. Nothing here is dispatched on an unmeasured size: the probe of §9 replaces every estimate first.

9 Next steps: the ladder — which stores are built first, second and third

phase	stores	stored	access
A	<code>ghcnd</code> , <code>igra</code> , <code>tide</code> and <code>tide_private</code> , <code>ndbc</code> , <code>gliders</code> , <code>flux</code> , <code>fire</code> , <code>lst05</code> (Terra), <code>snow05</code> (Terra), all ten <code>cat_*</code> catalogues (<code>cat_palsar</code> as soon as the JAXA account exists), <code>alerts</code> , <code>lossyear</code> , <code>burned500</code> , <code>cat_gfm</code> , <code>static_fine</code> , the registry; and the biosphere wave — <code>canopy30</code> , <code>canopy_ref</code> , <code>sif</code> (TROPOMI), <code>pheno500</code> , <code>gbif</code> , <code>cat_biomass_esa</code> , <code>biomass100</code> (2020 and the GEDI L4B group), one year of <code>gedi</code> and of <code>icesat2</code> , and <code>lai500</code> for Terra 2020–2024	≈ 0.7 TB, and ≈ 1 TB [†] more for the biosphere wave	seven need no account at all (<code>ghcnd</code> , <code>igra</code> , <code>tide</code> , <code>ndbc</code> , <code>gliders</code> , <code>lossyear</code> , <code>static_fine</code>); <code>flux</code> and <code>fire</code> ’s NOAA-20/21 archive go through a portal request; the NASA stores and catalogues need an Earthdata login (NASA’s free account) on a hosted runner (GitHub’s own build machines); the Sentinel catalogues a CDSE account; the Landsat one a USGS login or a requester-pays AWS read (a download billed to the reader); <code>lst05</code> is assembled on one verified box (a rented machine on a vetted host) (\$2–5); under \$10; <code>tide_private</code> is written by the <code>tide</code> build to the private repository (the project’s non-public Hugging Face dataset store). The biosphere wave adds no new credential except a free JAXA G-Portal account for <code>cat_palsar</code> : <code>canopy30</code> , <code>canopy_ref</code> , <code>gbif</code> and TROPOMI fluorescence are anonymous, the rest use the Earthdata login already held, and the OCO half of <code>sif</code> waits on the GES DISC approval
B	<code>ghcnh</code> , <code>hfr</code> , <code>river</code> , <code>river_grdc</code> , <code>ismn</code> , <code>alticap</code> , <code>refl05</code> , <code>ims1k</code> , <code>chirps05</code> , the Aqua and VIIRS continuations of <code>lst05</code> and <code>snow05</code> ; and what phase A measured before buying — the rest of <code>gedi</code> ’s and <code>icesat2</code> ’s records, the rest of <code>lai500</code> , <code>biomass100</code> ’s other years	2.3–2.7 TB, plus whatever the lasers’ probes say their records cost	\$20–30 in box time; the Hugging Face conversation about terabyte repositories first; <code>ismn</code> and <code>river_grdc</code> are registered downloads on a hosted runner, published to the private repository
C	<code>landcover10</code> , <code>ecad</code> , the tile tokens (the codec’s vectors per image tile) over E-078c’s coastal cells, <code>oc300_coast</code> (the 300 m colour of <code>cat_olci</code> stored over the band, ≈ 1.6 TB)	≈ 3 TB	after E-078c’s verdict

Table 7: The build ladder. Phase A is the eight point stores (≈ 150 GB, one of them private), the two Terra fields (≈ 410 GB), the catalogues and targets (≈ 80 GB), the statics, and the biosphere wave of \$4.3.

The biosphere wave, and what it builds first. Every store of \$4.3 is new, so nothing in it is dispatched before its own one-month probe has replaced this note’s estimates. Phase A then builds, in this order: the stores that are whole and cheap — `canopy30` (one year, 261 measured tiles), `pheno500` (the whole 2001–2025 calendar), `gbif` (the public records of one monthly snapshot), `sif` from TROPOMI, the two catalogues `canopy_ref` and `cat_biomass_esa`, and, in family 1.gf, the whole of `pace4k` (PACE’s daily 4 km ocean colour); then *one year* of each store whose record is a terabyte — `gedi`, `icesat2`, `lai500` (Terra 2020–2024) and `biomass100` (2020, with the whole GEDI L4B group beside it). The year is 2022 for the lasers because both flew it in full. What phase B buys is then a decision made on measured rows,

bytes and fetch rates rather than on an estimate.

The probe (a one-month test fetch through the real code). One month per source through the real adapter on a hosted runner: rows per day and distinct platforms; valid fraction and compressed bytes per tile; scene counts per day against the producer’s catalogue; throughput; and for `tide` the station-years the licence sort sends to `tide_private`.

Two falsifiers. For the *forcing and coastal* stores, family 1.gf’s test: at anchors where the store has coverage, its tokens in the cone (the past measurements the model reads around the predicted place, the anchor) must lower the held-out masked-token loss (the model’s error at predicting hidden tokens) on the goal’s read-out against the same anchors without them, three seeds (three independent training runs); parity means the reanalysis and the coarse tensor already carried it. For the *land targets* the test is reversed and is the family’s main one: the embedding trained on this family’s inputs must predict each target at its own footprint better than climatology (the long-term average for that place and season) and better than the same read-out from family 7.2 alone — next year’s alerts from this year’s state, biomass from imagery plus LiDAR, burned area from temperature and reflectance — and, for the forty-year record, the codec’s tokens of the same place from Landsat 5 and Landsat 8 must agree with each other more than with a different place, which is the test that the sensor shift was learned rather than memorised.

10 Decisions for Chris — open choices, each with this note’s pick

1. **The name:** 1.0.tf as given.
2. **The two tracks:** `ismn`, `river_grdc`, `ecad`, `tide_private` and NICFI on a private track rather than refused, so the soil-moisture store returns. Pick: yes, in the private repository of family 1.gf §4.
3. **The coastal band:** ± 100 km. Pick: 100 km.
4. **GEDI at full footprint resolution** (0.2–0.5 TB for the whole record) rather than thinned, with phase A building one year of it and the probe deciding the rest. Pick: full; thinning is derived.
5. **Landsat back to 1982 in the catalogue**, all sensors, HLS beside it. Pick: yes.
6. **Targets in phase A** (`alerts`, `lossyear`, `burned500`, `cat_gfm`), cheap and the reason the family exists. Pick: yes.
7. **Exceptions E4–E7:** CHIRPS as input; PALSAR mosaics and regional orthophotos in the catalogue; the eight-day leaf-area composite as input. Pick: admit all four; NICFI on the private track once its new-phase terms are read.
8. **Tide gauges:** GESLA-4 filtered per source, else UHSLC’s two sets. Pick: GESLA filtered.
9. **A JAXA G-Portal account** for `cat_palsar`, the yearly L-band radar mosaics — free registration on Japan’s space agency portal, and the one credential the biosphere wave needs that the project does not already hold. Pick: open the account; otherwise the store waits and the radar half of the biomass read-out is NISAR-only, which begins in 2026.
10. **The GES DISC approval** in the Earthdata profile — one click in Chris’s NASA account, which unblocks `xco2` and `irtb` in family 1.gf and the OCO-2 and OCO-3 half of `sif` here. Pick: approve; the TROPOMI half is built meanwhile and needs no account.
11. **GBIF’s non-commercial records:** build them on the private track as designed, or drop them. Measured on 2026-09-20 against GBIF’s own interface, of the records that carry coordinates and no geospatial problem, 11.0 % are CC0 and 72.2 % CC BY — 83.2 % public — and 16.8 % are CC BY-NC. Pick: the private track, because a sixth of the world’s species records is a sixth of the world’s species records; dropping them is the cheaper answer and costs coverage that is not evenly distributed.
12. **ESA CCI Biomass terms.** Read on 2026-09-20 at the producer’s own licence file: the data “may be used by any user for any purpose” with acknowledgement of the Climate Change Initiative and the project, and citation; rights stay with the producers; redistribution is neither granted nor forbidden in words. Pick: the public track, on the reading that a licence permitting any purpose with attribution permits republishing with attribution — but it is Chris’s call, and the private track costs nothing but reproducibility if he reads it the other way.

Terms used above, in one line each

Tier G / P / T family 10's three storage tiers: gridded fields; points, profiles and tracks; tiles of imagery held as a catalogue plus codec tokens. **STAC** SpatioTemporal Asset Catalog, the standard in which producers list scenes. **CSR** compressed sparse row — an offsets array saying where each five-day bin's rows start. **GSHHG** the Global Self-consistent Hierarchical High-resolution Geography shoreline set. **GHCNh / GHCN-Daily / ISD** NOAA's hourly and daily station archives and the frozen hourly archive GHCNh replaces. **IGRA** the Integrated Global Radiosonde Archive. **GESLA / UHSLC / CO-OPS** the global hourly tide-gauge compilation, the University of Hawaii Sea Level Center, NOAA's tide-gauge service; **CV, UZ, CMEMS** three GESLA source codes with research-only terms; **SANHO** the South African Navy Hydrographic Office. **NDBC / C-MAN** NOAA's buoy centre and its coastal automated stations. **HFRNet / IOOS** the US high-frequency coastal radar network and the Integrated Ocean Observing System, which also runs the glider data-assembly centre (DAC). **FLUXNET / AmeriFlux / ICOS** networks of eddy-covariance towers measuring the exchange of CO₂, water and heat between land and air; **NEE / GPP / Reco / LE / H / Rn / VPD / SWC** net ecosystem exchange, gross primary production, ecosystem respiration, latent and sensible heat, net radiation, vapour-pressure deficit, soil water content. **FIRMS / FRP** NASA's fire-detection service and fire radiative power. **NWIS** the US Geological Survey's water archive. **ALTICAP / ODATIS / AVISO** a coastal altimetry product and the French ocean-data services that host it. **GEDI / ICESat-2 / ATLAS / ATL08** NASA's two spaceborne LiDARs, the ICESat-2 instrument, and its land-and-canopy product; **RH_n** the relative height below which *n*% of a LiDAR footprint's returned energy lies — RH98 is the canopy top. **CMG** MODIS' Climate Modeling Grid, 0.05°. **IMS** the US National Ice Center's daily snow and ice chart. **CHIRPS** the Climate Hazards infrared precipitation with stations. **L1B / L2 / L2A / L3 / GRD / SLC / GCOV** calibrated radiances; retrievals; Sentinel-2's surface reflectance; gridded; Sentinel-1's detected and complex radar products; NISAR's geocoded covariance. **TM / ETM+ / OLI / TIRS** the Landsat 4–5, 7, and 8–9 instruments. **HLS** Harmonized Landsat Sentinel-2. **PALSAR / NISAR** JAXA's and NASA-ISRO's L-band radars. **OLCI WFR / VIIRS / ECOSTRESS** Sentinel-3's ocean-colour product, the JPSS imager, and the ISS thermal imager. **NAIP / NICFI** the US aerial orthophoto programme and Norway's tropical-forest satellite programme. **GLAD / RADD / DIST-ALERT** forest-disturbance alert systems from the University of Maryland, Wageningen, and NASA OPERA. **JRC TMF** the EU Joint Research Centre's tropical moist forest maps. **MCD64A1** MODIS burned area. **GFM / DSWx** Copernicus' flood monitoring and OPERA's surface-water maps. **ESA CCI** ESA's Climate Change Initiative. **GHSL / WSF** the Global Human Settlement Layer and the World Settlement Footprint. **CDL / EUCROPMAP / WorldCereal** crop-type maps from the USDA, the JRC and ESA. **ISMN / ECA&D / GRDC** the soil-moisture network, the European station set and the Global Runoff Data Centre — private-track stores; **HadISD / MSWEP** the Met Office hourly station set and a merged precipitation product, set aside as derivatives; **Caravan** GRDC's CC BY subset, public. **Public / private track** a store anyone may read from the public repository, or one held for this project's own use because its licence forbids redistribution (family 1.gf §4). **GEBCO** the General Bathymetric Chart of the Oceans. **E-078 / E-078c / E-081** the multi-granularity design, its first coastal tile-codec experiment, and the fishing-effort store. **AMOC / SST** the Atlantic Meridional Overturning Circulation and sea-surface temperature. **CC0 / CC BY / CC BY-NC** public-domain dedication, attribution, and non-commercial Creative Commons licences. **MODIS / VIIRS** the imaging radiometers on Terra and Aqua, and on Suomi-NPP, NOAA-20 and NOAA-21. **LP DAAC / NSIDC / EOSDIS** NASA's land-processes and snow-and-ice data centres and the system that serves them. **CDSE** the Copernicus Data Space Ecosystem, ESA's Sentinel archive. **ERDDAP** a data server that answers tabular queries over HTTP. **LSTE** land-surface temperature and emissivity. **GRDH** ground-range-detected high-resolution. **GLO-30** the Copernicus 30 m elevation model. **OGD** open government data. **TERN / ChinaFLUX** the Australian and Chinese flux-tower networks. **Dynamic World** Google's 10 m per-scene land-cover classification, Earth Engine only. **NaN** not-a-number, the stored value for “not measured”. **SIF** solar-induced chlorophyll fluorescence, the faint red light a leaf re-emits while photosynthesising — the closest thing to a direct satellite measurement of photosynthesis. **TROPOMI / S5P-PAL / TROPoSIF** the trace-gas imager on Sentinel-5P, the ESA service that publishes extra products from it, and its fluorescence product. **OCO-2 / OCO-3 / SIF Lite** NASA's two Orbiting Carbon Observatories and their per-sounding fluorescence files. **GES DISC** NASA's atmosphere data centre, which serves the OCO products and needs its own approval on an Earthdata account. **LAI / FPAR** leaf-area index, the area of leaf above a square metre of ground, and the fraction of photosynthetically active radiation a canopy absorbs. **MOD15A2H / MYD15A2H / VNP15A2H** the eight-day leaf-area and FPAR products from Terra, Aqua and VIIRS. **MCD12Q2 / EVI** MODIS' yearly land-surface phenology product, and the enhanced vegetation index its dates are derived from. **Phenology** the calendar of a plant's year: green-up, peak, senescence, dormancy. **GLAD** the University of Maryland's Global Land Analysis and Discovery laboratory, producer of the Hansen loss maps, the alerts and the 2020 canopy-height map. **GBIF** the Global Biodiversity Information Facility, the pooled archive of species-occurrence records; **taxon key** its identifier for a species or higher group; **basis of record** how an occurrence was observed (specimen, human observation, machine). **ETH Zurich / Meta-WRI canopy height** the 10 m and 1 m canopy-height maps of other groups, catalogued rather than copied. **BIOMASS / P band** ESA's 2025 radar satellite and its 70 cm wavelength, which reaches through a canopy to the trunks. **PACE / OCI** NASA's 2024 ocean-colour satellite and its instrument; its daily 4 km store **pace4k** is family 1.gf's. **E7** the exception admitting the eight-day leaf-area composite (§2). **Parquet** a columnar table file format, the form GBIF publishes in.

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